

FMCG Sales Performance Dashboard Report

Introduction

The FMCG Sales Performance Dashboard is an interactive Power BI project developed to analyze sales trends, brand performance, promotional effectiveness, channel contribution, and delivery performance. The dashboard helps business stakeholders make data-driven decisions by providing clear visual insights into key business metrics.

1. Project Objectives

- ✓ Analyze overall sales performance.
- ✓ Monitor monthly revenue trends.
- ✓ Identify top-performing brands.
- ✓ Evaluate the effectiveness of promotional campaigns.
- ✓ Compare sales across different channels and regions.
- ✓ Track delivery performance and operational efficiency.

2. Dataset Description

The dataset contains FMCG sales information including:

- Date
- Brand
- Category
- Region
- Channel(RETAIL,E-COMMERCE,DISCOUNT)
- Revenue
- Units Sold
- Delivered Quantity
- Delivery Days
- Promotion Flag

3. Data Cleaning and Preprocessing

Before developing the dashboard, the dataset was cleaned and preprocessed using Google Colab with Python libraries such as Pandas and NumPy. The objective was to ensure data quality and prepare the dataset for accurate analysis and visualization in Power BI.

The following preprocessing steps were carried out:

- Imported the dataset into Google Colab.
- Examined the dataset dimensions, structure, and data types.
- Generated summary statistics to understand the distribution of numerical variables.
- Checked for missing values and confirmed that no missing values were present in the dataset.
- Checked for duplicate records and confirmed that no duplicate records were present.
- Verified data types and converted date columns into the appropriate datetime format.
- Performed data validation checks on numerical columns.
- Identified invalid negative values in stock availability, delivered quantity, and units sold.
- Removed invalid records to maintain data consistency and business accuracy.
- Exported the cleaned dataset for visualization and dashboard development in Power BI.

These preprocessing activities ensured that the dataset was clean, reliable, and ready for business intelligence reporting.

Data Cleaning Workflow

Figure 1: Dataset Overview in Google Colab

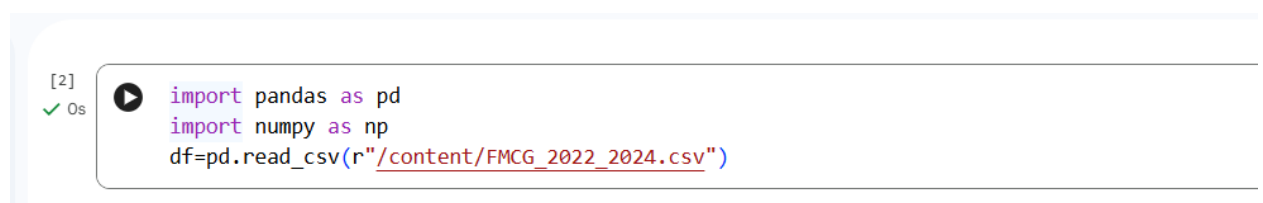


Figure 2: Dataset Structure and Summary Statistics

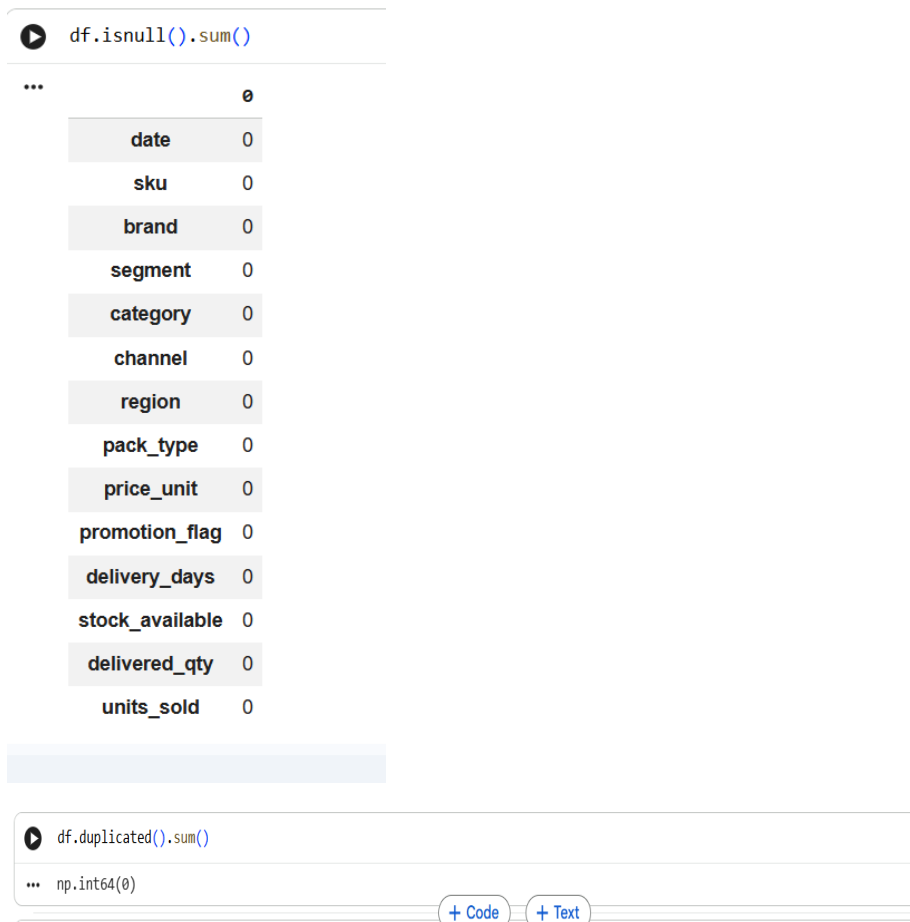


Figure 3: Detection of Invalid Negative Values

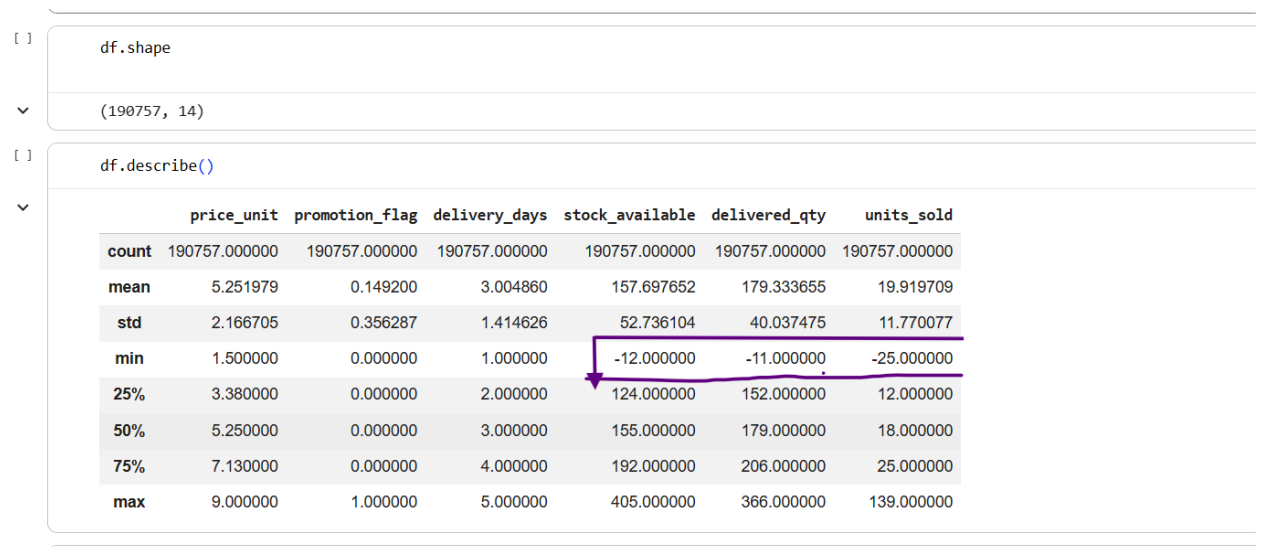


Figure 4: Data Cleaning and Validation Process

```
numerical_df = df.select_dtypes(include=np.number)
df[numerical_df.columns] = numerical_df.mask(numerical_df < 0, 0)
```

```
df.describe()
```

	price_unit	promotion_flag	delivery_days	stock_available	delivered_qty	units_sold
count	190757.000000	190757.000000	190757.000000	190757.000000	190757.000000	190757.000000
mean	5.251979	0.149200	3.004860	157.697757	179.333765	19.919898
std	2.166705	0.356287	1.414626	52.735781	40.036970	11.769602
min	1.500000	0.000000	1.000000	0.000000	0.000000	0.000000
25%	3.380000	0.000000	2.000000	124.000000	152.000000	12.000000
50%	5.250000	0.000000	3.000000	155.000000	179.000000	18.000000
75%	7.130000	0.000000	4.000000	192.000000	206.000000	25.000000
max	9.000000	1.000000	5.000000	405.000000	366.000000	139.000000

Figure 5: Cleaned Dataset Ready for Power BI

```
df.to_csv("FMCG_Cleaned.csv", index=False)
from google.colab import files

files.download("FMCG_Cleaned.csv")
```

4. Dashboard Overview

The cleaned dataset was imported into Power BI to create an interactive dashboard for analyzing FMCG sales performance. The dashboard was designed across two pages to provide both high-level and detailed business insights.

Page 1: Sales Overview

This page provides a summary of overall sales performance and allows users to analyze sales across different dimensions.

Visualizations Included

Year Slicer

Allows users to filter dashboard insights based on the selected year.

Region-wise Sales Analysis

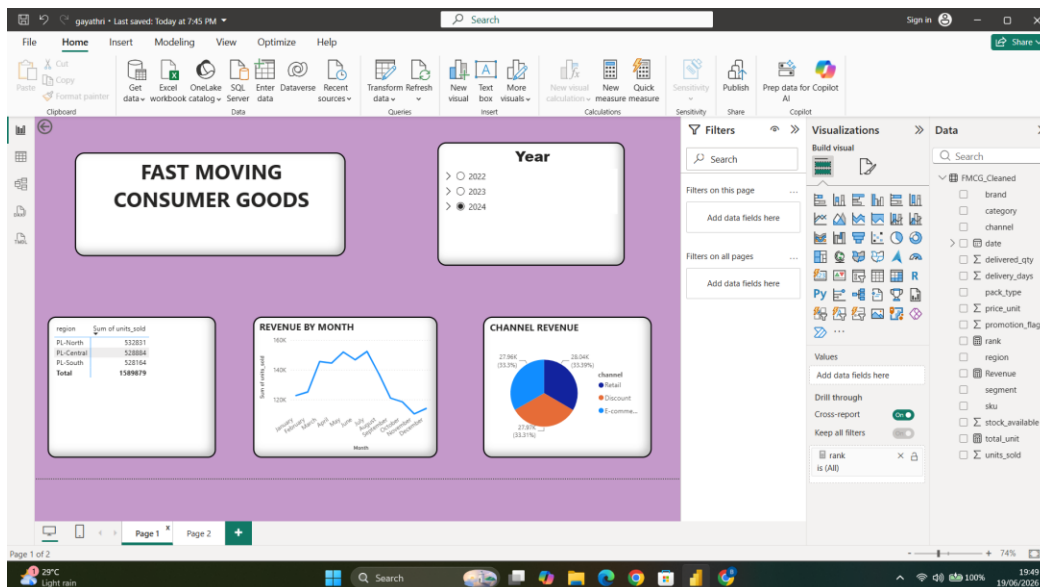
Displays total units sold across different regions, enabling comparison of regional performance.

Revenue by Month

A line chart showing monthly sales trends and helping identify seasonal patterns in sales performance.

Channel Revenue Distribution

A pie chart illustrating the contribution of different sales channels such as Retail, Discount, and E-commerce.



Page 2: Performance Analysis

This page focuses on key performance indicators, brand performance, and promotional effectiveness.

Visualizations Included

Total Revenue KPI

Displays the total revenue generated from sales transactions.

Average Delivery Days KPI

Shows the average number of days required to deliver products, helping evaluate operational efficiency.

Top 5 Brands

Highlights the best-performing brands based on sales volume.

Sales by Promotion Status

Compares sales performance between promoted and non-promoted products.

Pack Type Analysis

Analyzes sales distribution across different packaging types such as Single, Carton, and Multipack.



5. Key Insights

- Total Revenue reached 19.95M.
- Average delivery time is approximately 3 days.
- YoBrand4 is the top-performing brand.
- Non-promotional sales contribute a larger share of total sales.
- Multipack and Single pack products show strong sales performance.

6. Business Recommendations

Based on the analysis of sales performance, delivery metrics, brand contribution, and promotional activities, the following recommendations are suggested:

1. Focus on Top-Performing Brands

Brands such as YoBrand4 and YoBrand3 contribute significantly to overall sales. Increasing inventory availability and marketing efforts for these brands can further improve revenue.

2. Optimize Underperforming Brands

Brands with lower sales performance should be reviewed to identify factors affecting demand. Promotional campaigns and product repositioning strategies may help improve sales.

3. Strengthen Promotion Strategies

The analysis indicates that promotions influence sales performance. The company should continue targeted promotional campaigns for products and regions with growth potential.

4. Improve Regional Sales Performance

Sales performance varies across regions. Regions with lower sales should be targeted with localized marketing strategies and improved product availability.

5. Maintain Delivery Efficiency

The average delivery time is approximately 3 days, indicating efficient logistics operations. Maintaining this performance can contribute to customer satisfaction and repeat purchases.

6. Utilize Sales Trends for Forecasting

Monthly sales patterns can be used for demand forecasting and inventory planning, helping reduce stock shortages and excess inventory.

7. Monitor Channel Performance

Regular evaluation of Retail, Discount, and E-commerce channels can help allocate resources more effectively and maximize revenue generation.

7. Conclusion

The FMCG Sales Performance Dashboard successfully transformed raw sales data into actionable business insights through interactive visualizations and key performance indicators. The analysis highlighted important trends related to revenue, brand performance, promotional effectiveness, sales channels, and delivery efficiency.

The dashboard enables stakeholders to make informed decisions, monitor business performance, and identify opportunities for growth. By implementing the recommended strategies, organizations can improve operational efficiency, enhance customer satisfaction, and achieve better business outcomes.